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Article

Quantifying Indirect Contribution from Precursors to Human Body Burden of Legacy PFASs Based on Paired Blood and One-Week Duplicate Diet

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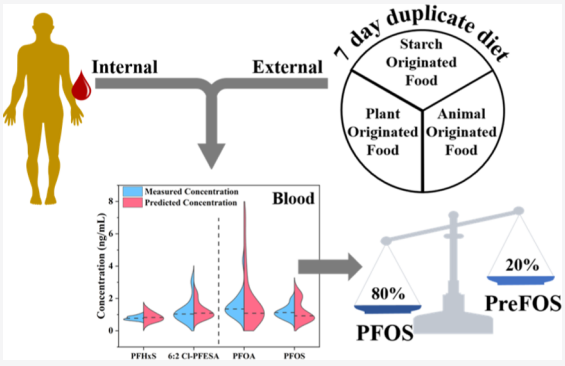
ABSTRACT: The restriction on legacy perfluoroalkyl substances (PFASs) has led to increasing application and contamination of their precursors and novel alternatives. However, the indirect contribution from precursors has not been well characterized. In this study, 24 PFASs were measured in the paired human blood and urine from general volunteers (n = 20), as well as their corresponding exposure matrices (7 day duplicate diet, drinking water and dust). Perfluorooctanoic acid (PFOA) was predominant, followed by 6:2 chlorinated polyfluoroalkyl ether sulfonate (6:2 Cl-PFESA), contributing 21.6−47.0 and 6.6−20.0% of the total concentrations, respectively. Total oxidable precursor (TOP) assay and isomeric analysis coupled with a toxicokinetic model suggested that around 19% of perfluorooctane sulfonate (PFOS) in human was contributed by its precursors. The strong correlation between the estimated daily intake (EDI) and human blood concentration for 6:2 Cl-PFESA suggested that it was mainly contributed by direct exposure. The bioavailability of 6:2 Cl-PFESA in the food matrices was estimated as 18.6% by comparing the estimated and measured blood concentrations, implying that human exposure might be overestimated if the bioavailability of PFASs in food was not considered. Assuming that they had a similar bioavailability, it was estimated that ca. 20% of PFOS body burden was from indirect exposure to its precursors, which was supported by TOP assay.



KEYWORDS: *PFASs, isomer profiles, human exposure, bioavailability, indirect exposure*

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■ INTRODUCTION

Per- and polyfluoroalkyl substances (PFASs) are a big family of artificial fluoro-substituted chemicals that have been widely used in industrial processes and commercial products over the last 70 years.[1,2] Due to great persistence and strong bioaccumulation potentials, they are ubiquitous in the environment, organisms, and humans and had caused various toxicological effects on animals and humans.[1,3] To tackle down their worldwide pollution, two of the most typical homologues of perfluoroalkyl acids (PFAAs), perfluorooctane sulfonate (PFOS) and perfluorooctanoic acid (PFOA), were included in the list of persistent organic pollutants, while perfluorohexane sulfonate (PFHxS) was under scrutiny.[4] Thus, the production and application of these legacy PFAAs are restricted and gradually phased out from the global market.[2,5] In order to make up for market demand, a range of alternatives, such as chlorinated polyfluoroalkyl ether sulfonates (mainly 6:2 Cl-PFESA), and precursors, including fluorotelomer sulfonic acids (FTSs, including 6:2 and 8:2 FTS), fluorotelomer phosphate diesters (diPAPs, including 6:2 and 8:2 diPAP), and perfluorooctane sulfonamide (PFOSA), were developed and are extensively applied in the world.[6,7] Many of them may be transformed to PFOS or PFOA in the environment and organisms, and thus, they are referred as PFOS or PFOA precursors.[2]

Evaluation of the relative contributions of direct and indirect (via precursors) exposure is meaningful for human health risk assessment and pollution control. The isomer profiling technique provides useful information to track the relative contributions by comparing the isomeric proportions of PFASs in organisms and in industrial products.[8] It is well known that perfluorosulfonate acids (PFSAs) are generally manufactured by electrochemical fluorination (ECF), which yields a range of carbon-chain skeleton isomers (e.g., for PFOS, ca. 70% linear isomer and ca. 30% branched isomers).[8] In wild organisms, n-PFOS with higher accumulation potential usually contributed more than 80% of total PFOS.[9,10] In contrast, n-PFOS generally comprised only 45−67% of total PFOS in general

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population blood.[9,11,12] The lower percentage of *n*-PFOS in human blood was supposed to be related to the indirect exposure sources of PFOS precursors, even though no solid evidence has been found.[13] In most of these studies, the blood PFOS isomeric profiles were compared directly with the ECF products, but not with human exposure matrices, which would lead to misinterpretation of human exposure results. Additionally, the judgements were usually qualitative without solid quantitative assessment.

Estimated daily intake (EDI) is commonly used to characterize human exposure and associated health risks of PFASs by comparing with the established criteria or threshold values.[14−17] In most of the surveys, the EDIs were estimated based on raw food materials and general food consumption data without considering the impact of cooking on dietary intake of PFASs, which could be misleading.[18] Some studies reported that the concentrations of short-chained PFASs decreased, while long-chained PFASs increased during cooking, such as frying, steaming, or boiling.[18,19] Recently, a 1 or 2 day duplicate diet was employed to assess EDI, which was proven to improve human exposure assessment.[20,21] However, the 1 or 2 day duplicate diet does not well reflect variations in dietary intakes of the participants because most people tend to change their food recipes within a short period (such as a week). On the other hand, there are many studies reporting the internal exposure of PFASs by measuring them in human blood[12,22] and attempted to understand the intrinsic relationship between external and internal exposures by a one-compartment toxicokinetic model, in which bioavailability of PFASs in the food matrices was assumed to be 100%.[23−26] However, some recent studies suggested that the bioavailability was lower than 100% in many circumstances, implying that the human exposure risks might be overestimated in previous studies.[23,24,27]

Hence, to better characterize the indirect and direct exposure sources of PFASs among general population, 20 volunteers were recruited, and the corresponding blood, urine, and exposure matrices (1 week duplicate food, drinking water, and indoor dust) of each participant were collected. The objectives of this work are to (1) quantify PFASs and the isomers of PFOS, PFOA, and PFHxS in the abovementioned samples; (2) estimate bioavailability of PFASs in the food matrices by comparing the measured and predicted concentrations of PFASs in the blood; and (3) characterize the contribution of indirect exposure to the PFAS body burden in humans. Multiple methods including isomeric profiling technique, total oxidable precursor (TOP) assay, and one-compartment pharmacokinetic model were carried out. The findings of this study may improve the understanding of human exposure to PFASs and distinguish the contribution of indirect exposure.

■ MATERIALS AND METHODS

Chemicals and Reagents. The following compounds were analyzed in samples: legacy C_{4-13} perfluorocarboxylic acids (PFCAs) and $C_{4,6,8,10}$ PFSAs (including isomers of PFOA, PFOS, and PFHxS), four precursors of PFOS (PFOSA and its N-substituted derivatives), four precursors of PFCAs (6:2 and 8:2 diPAP, 6:2 and 8:2 FTS), and two novel PFASs (6:2 and 8:2 Cl-PFESA). All chemical standards and corresponding internal standards were purchased from Wellington Laboratories (Ontario, Canada). The detailed information of standards, along with nomenclature and compositions for PFOA, PFOS, and PFHxS isomers, is provided in the Supporting Information (Table S1).

Participants and Sample Collection. Twenty volunteers (14 males and 6 females) were recruited in Tianjin, China, with the age of 20−25. All the research protocols were approved by the Ethics Committee of Nankai University, China. Before sampling, a 30 day food recipe questionnaire for each subject was recorded, which showed that their food recipes did not change much among weeks and each volunteer had fixed drinking water sources. Thus, for a continuous week (7 days) in October 2019, each subject was required to provide duplicates of the three meals they actually ate on each day. The food items were then classified into starch-originated food (SOF, including rice- and flour-based products), plant-originated food (POF, including a variety of vegetables and soy products), and animal-originated food (AOF, including meat, eggs, fish, and seafood). Thus, there was a total of 420 food samples, most of which were cooked food. Simultaneously, drinking water samples (n = 20), which were mainly bottled water and purified water, and indoor dust (n = 20) were provided by the participants. At the last day of sampling, whole blood and morning urine (n = 20) were collected from each subject. Detailed information about the sampling design could be found in the Supporting Information.

Extraction and Clean up. An aliquot of 15 g (wet weight) of the food sample was homogenized and freeze-dried. After being spiked with internal standards (1 ng of each), ion-pair extraction was conducted followed by SPE purification on the WAX cartridge (500 mg/6 mL, Agela, Tianjin, China) according to a previous study.[28] Drinking water, indoor dust, blood, and urine samples were pretreated according to the literature methods.[25,29−31] The indoor dust (200 mg, 1 ng internal standard of each compound, the same below) and blood (1 mL) samples were extracted with organic solvents and further cleaned by PestiCarb cartridges (200 mg/6 mL, Agela, Tianjin, China). The drinking water (500 mL) and urine (50 mL) samples were loaded onto WAX cartridges (500 mg/6 mL and 250 mg/6 mL, respectively) for enrichment and purification. Detailed information of pretreatments is provided in the Supporting Information.

Instrumental Analysis. A total of 24 PFASs, including the isomers of PFOS, PFOA, and PFHxS, were measured on an Acquity UPLC coupled to a Xevo TQ-S triple quadrupole mass spectrometer (Waters, MA, USA). A Waters UPLC BEH C18 column was used to separate the novel and legacy PFASs, and a Supelco Ascentis Express (Supelco, PA, USA) column was used for isomer separation. Detailed information for the instrumental analysis is provided in the Supporting Information (Table S2).

TOP Assay. TOP analysis was applied for identifying unknown precursors of legacy PFASs in the exposure sources, during which the precursors could be transformed to legacy PFAAs.[32] The dietary samples of each subject were pooled proportionally as one sample (10 g) according to the collected wet weights. According to Houtz and Sedlak,[32] after extraction, the sample was reconstituted with water (pH > 12), to which potassium persulfate was added. Then, the mixture was incubated in a water bath at 85 °C for 6 h. Afterward, the pH of the sample was adjusted to 7.0 and internal standards were added. The samples were then loaded onto a WAX cartridge for enrichment. Detailed information of TOP assay can be found in the Supporting Information.

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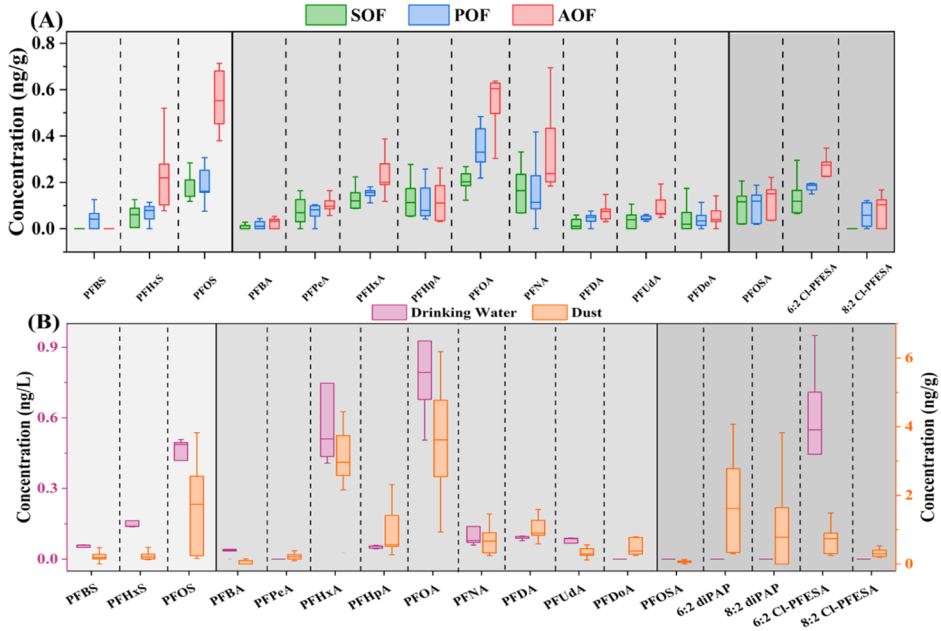

Figure 1. Concentrations of PFASs in the (A) three kinds of foods and (B) drinking water and indoor dust.

Renal Clearance (CL) Estimation. Daily CL (mL/day/kg) is defined as the volume of blood from which a chemical is completely removed by urinary excretion in 1 day per body weight (eq 1).[31]

$$\mathrm{CL} = \frac{C_{\mathrm{urine}} \times V_{\mathrm{urine}}}{C_{\mathrm{blood}} \times \mathrm{BW}} \tag{1}$$

C_{urine} and C_{blood} are the concentrations of a certain PFAS in urine (ng/L) and blood (ng/mL), respectively, V_{urine} is the daily urine excretion volume (L), which is estimated to be 1.2 and 1.4 L for female and male, respectively,[33] and BW is the individual body weight of a specific participant (kg).

Estimated Daily Intake (EDI) Calculation. The exposure through dermal absorption and hand-to-mouth contact was neglected due to their considerably low contribution for adults.[21,24] Thus, the direct exposure of a special PFAS to humans is mainly through consumption of food, drinking water, dust ingestion, and air inhalation. Thus, total EDI ($\mathrm{EDI}_{\mathrm{total}}$, ng/kg/day) of PFASs was the sum of the four items, which was formulated in the Supporting Information.

Modeling the Blood Concentration of PFASs. A one-compartment pharmacokinetic model was used to predict the human blood levels of PFASs and isomeric proportions of PFHxS, PFOA, and PFOS based on the EDI using the following equation, assuming PFASs achieved the steady state in the participants

$$C_{\mathrm{M}} = \frac{\mathrm{EDI} \times F}{k_{\mathrm{e}} \times V_{\mathrm{d}}} = \frac{\mathrm{EDI} \times F \times 365 \times t_{1/2}}{V_{\mathrm{d}} \times \ln 2} \tag{2}$$

C_{M} is the modeled blood concentration (ng/mL) and k_{e} is the first-order elimination rate, defined as ln2 divided by elimination half-life ($t_{1/2}$) in years. The half-lives of blood elimination PFOA, PFOS, and PFHxS (1.7, 1.9, and 3.6 years, respectively) were obtained from the study by Fu et al.[34] V_{d} is the volume of distribution, and the values for PFOA, PFOS, and PFHxS were set to be 170, 230, and 213 mL/kg, respectively.[26] Due to the limited toxicological studies on 6:2 Cl-PFESA, the $t_{1/2}$ and V_{d} of 6:2 Cl-PFESA were assumed to be the same as PFHxS because of their similar CL values in this study (which will be shown in the results and discussion part). F refers to the bioavailability of PFASs in food.

Quality Assurance and Control. A procedural blank was included with each batch of 10 samples, and no contamination was found. A seven-point calibration curve of each PFAS compound with an internal standard was established for quantification ($R^2 \geq 0.99$). Limit of detection (LOD) and limit of quantification (LOQ) were defined as the concentrations with a signal-to-noise ratio of three and ten. The concentrations lower than LODs were defined as 0, while those lower than LOQs were defined as corresponding LOQ values divided by the square root of two. The recoveries of most of the targeted compounds were 63.3−130% with a standard deviation of 2.46−13.1%. Detailed information of recoveries and LOQs is available in the Supporting Information (Table S3).

Statistical Analysis. Nonparametric statistical methods were applied because most of the concentrations of PFASs including isomers were not normally distributed even after log transformation. For the statistical purpose, certain PFASs were excluded if their detection frequencies (DFs) were <50%. The Mann−Whitney−Wilcoxon test was conducted to compare PFAS concentrations and isomer proportions in different matrices. Spearman rank correlation was performed to explore the relationships between the PFAS concentrations in exposure matrices (or urine) and blood. All statistical analyses were performed using R (version 4.0.2).

■ RESULTS AND DISCUSSION

PFASs in the Human Exposure Matrices. *Diet.* Most of legacy PFASs, while only a few precursors and novel PFASs were detected in the three diet categories (Table S4). As shown in Figure 1A, in AOF, PFOA (0.84 ng/g, mean, the same as follows) was followed by PFOS (0.82 ng/g), PFNA (0.36 ng/g), and PFHxA (0.23 ng/g). Similarly, in the POF, PFOA (0.58 ng/g) was followed by PFOS (0.21 ng/g) and PFHxA (0.16 ng/g), while in the SOF, PFOA (0.31 ng/g) was

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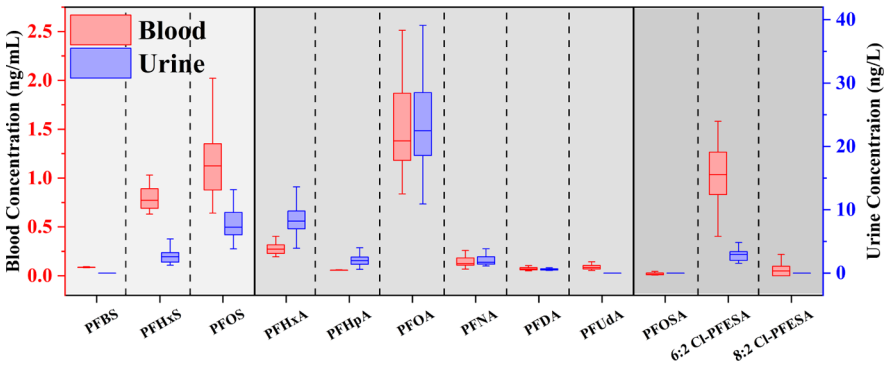

Figure 2. Concentrations of PFASs in the whole blood and urine.

followed by PFNA (0.22 ng/g), PFOS (0.19 ng/g), and PFHxA (0.13 ng/g).

As for the emerging species and precursors, only 6:2 Cl-PFESA and PFOSA were measurable, with the mean concentration in the ranges of 0.13−0.23 and 0.09−0.12 ng/g, respectively. The vigorous cooking processes might lead to the transformation of upper-stream precursors [such as *N*-ethylperfluoro-1-octane sulfonamide (*N*-EtFOSA)] to PFOSA and then to PFOS.[18,35] As a consequence, the precursors were barely detected except for PFOSA. Overall, the concentrations of PFASs in the diets were comparable to those reported in the studies conducted in the recent 5 years, while higher than those sampled before 2010, which might be due to strict restriction on PFOS in China that was implemented in 2019.[16,28,36−38]

Drinking Water. Similar to the food samples, PFOA (0.79 ng/L) made the highest contribution of 27.4% to the ΣPFASs. However, as shown in Figure 1B, 6:2 Cl-PFESA (0.59 ng/L) and PFHxA (0.58 ng/L) were the second prominent, followed by PFOS (0.46 ng/L), accounting for 19.8, 19.4, and 15.4% to ΣPFASs, respectively (Table S5). The detected concentrations of PFASs were generally lower than previous reports in China, possibly due to the fact that the drinking water samples collected were not tap water but purified or bottled water (Table S6).[25,39] It is quite surprising that 6:2 Cl-PFESA was present in the drinking water at a level slightly higher than that of PFOS. Although Cl-PFESAs have been detected in river water, lake water, and even marine water,[40,41] this is the first observation of its presence in drinking water, implying its prevalent contamination and potential human exposure.[42]

Indoor Dust. Akin to the food samples, PFOA (3.66 ng/g) was also predominant in the indoor dust with a contribution of 23.7% to ΣPFASs, which was followed by PFHxA and PFOS, comprising 19.4% (2.96 ng/g) and 11.4% (1.74 ng/g), respectively (Figure 1B and Table S7). Among the detected precursors and emerging PFASs, 6:2 diPAP (1.62 ng/g) and 8:2 diPAP (0.78 ng/g) (which was comparable to the previous study[29]) and 6:2 Cl-PFESA (0.74 ng/g) and 8:2 Cl-PFESA (0.29 ng/g), contributed 10.6, 5.1, 4.85, and 1.9% to the ΣPFASs, respectively.

EDI. For individual participants, the EDIs of the six main PFASs through direct exposure were calculated (see Table S8). The EDI_{total} of PFOA was the highest, with the mean value of 5.64 ng/kg/day, followed by PFOS (3.37 ng/kg/day), PFNA (2.73 ng/kg/day), PFHxA (1.82 ng/kg/day), 6:2 Cl-PFESA (1.81 ng/kg/day), and PFHxS (1.07 ng/kg/day).

The EDI_{air} values of PFOA and PFOS (0.031 and 0.047 ng/kg/day, respectively[29]) were two magnitudes lower than the corresponding EDI_{total}. Thus, the EDI_{air} was excluded and not considered anymore in the following discussion. In regard to other three exposure sources, dietary intake was predominant by contributing ≥96% of the EDI_{total} of these PFASs, which was in agreement with previous reports.[21] In general, the calculated EDI_{total} values of PFOA, PFOS, and PFHxS were slightly higher than those reported in European countries at <1.2, <1.1, and <0.2 ng/kg/day, respectively,[24,43] while comparable to two recent studies on Chinese general population, where the mean dietary exposures were <3 and <4 ng/kg/day for PFOA and PFOS, respectively.[28,44]

PFASs in Human Whole Blood and Urine. Legacy $C_{4,6,8}$-PFSAs and C_{6-11}-PFCAs were detected in the human blood samples with a DF of 100%, while others were occasionally detected (DF < 15%). Among the precursors and novel PFASs, only PFOSA (DF 100%), 6:2 Cl-PFESA (100%), 8:2 Cl-PFESA (55%), and 6:2 diPAP (20%) were detected in the blood samples (Table S9).

As shown in Figure 2, PFOA, PFOS, 6:2 Cl-PFESA, PFHxS, and PFHxA were the dominant species in the human blood. PFOA (1.61 ng/mL) made the most dominant contribution (28.5%) to ΣPFASs (5.65 ng/mL), followed by 6:2 Cl-PFESA (1.23 ng/mL, 21.8%), PFOS (1.16 ng/mL, 20.5%), PFHxS (0.77 ng/mL, 14.0%), and PFHxA (0.28 ng/mL, 5.0%). Taking the most prominent PFOA and PFOS as indicators, although their concentrations in this study were slightly lower than those in previous studies (ca. 1 vs several ng/mL), they did not show an obvious trend of decline like in the Western countries.[45] The mean concentration of 6:2 Cl-PFESA (1.23 ng/mL) in the blood was slightly lower, but also at the comparable level as those (several ng/mL) reported in recent years for general population in Tianjin and other Chinese cities.[12,22,46] There were significantly positive correlations ($p < 0.05$) between PFOS and 6:2 Cl-PFESA, PFOSA, and C_{8-10} PFCAs, as well as between PFOA and PFOS, PFNA, and PFDA (Figure S1), implying that these compounds had common exposure sources or pathways in humans. The concentrations of PFHxS and 6:2 Cl-PFESA in the male were significantly higher than those in the female ($p < 0.05$), which was in line with a previous study and might be due to their specific elimination via menstruation in females.[47]

In the human urine samples, only PFHxS, PFOS, C_{6-10}-PFAAs, and 6:2 Cl-PFESA were detected with a DF of 100%, much lower than those in the blood, while concentrations were significantly correlated for most PFASs (Tables S10−S11). PFOA (22.5 ng/L) made much more contribution (47.0%) to the ΣPFASs, followed by PFHxA (8.67 ng/L, 16.6%), PFOS (7.72 ng/L, 14.8%), 6:2 Cl-PFESA (3.43 ng/L, 6.6%), and

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PFHxS (2.69 ng/L, 5.2%). The CL values of the major PFASs were calculated based on the paired blood−urine samples (Table S12), and they were comparable with those reported previously (Table S13). The CL values decreased as the carbon chain length increased for PFCAs. For the three compounds with a sulfonate end group, the CL values of PFHxS (0.07 mL/day/kg) and 6:2 Cl-PFESA (0.07 mL/day/kg) were lower than that of PFOS (0.14 mL/day/kg), and all were lower than that of PFOA. This result indicated that the two PFOS alternatives might have slower urinal elimination potential than PFOS and thus have longer half-live in humans.[48,49] The CL value of PFHxS was in line with previous reports, while that of 6:2 Cl-PFESA was a magnitude higher than a previous study.[31,48,50]

Isomer Profiles of PFHxS, PFOA and, PFOS in Pooled Food, Blood, and Urine. Because food consumption was responsible for more than 96% of PFAS intake, the isomeric profiles in the composite food (which was pooled based on the collected wet weight of food) could represent the isomeric profiles in the exposure sources. The main isomers of PFHxS, PFOA, and PFOS were frequently detected in the pooled food, blood, and urine samples (Figure 3 and Tables S14−S16). The isomeric profiles of PFOA and PFOS were similar to the results reported previously on the Tianjin residents in 2012,[8] which did not exhibit an obvious temporal decline of % *n*-isomer in European countries.[51]

For the first time, we reported the compositions of five monomethyl substituted isomers of PFHxS in the pooled food, blood and urine samples. Except that 1*m*-PFHxS was not detected in the urine, other isomers were detected in all kinds of samples. The *n*-PFHxS was consistently the predominant isomer in the pooled food, blood, and urine samples, comprising 88.6, 90.6, and 89.6% of the ∑PFHxS, respectively, followed by *iso*- (4.5, 5.1, and 5.6%) and other branched (*br*-) isomers (≤4.0% for each), respectively.[12,52] The percentage of *n*-PFHxS in the food was similar to that in blood.

The isomer profile of the PFHxS industrial product (90.4% *n*-, and 5.5% *iso*-) is different from that of PFHxS (83.2% *n*- and 10.1% *iso*-) as an impurity in the PFOS industrial product (10−17%).[53] The isomer profile of PFHxS in the pooled food sample was greatly different from that in the PFOS commercial product as impurity ($p < 0.05$), while it was similar to that in the PFHxS industrial product (Table S17). In addition, the mean concentration of PFHxS (0.09 ng/g) was 1−4 times lower than that of PFOS (0.29 ng/g) in the diet. These together suggested that PFHxS in the exposure sources and participants was mainly attributed to the direct application of PFHxS (as a substitute of PFOS).

The *n*-PFHxS was enriched in the human blood relative to the food and urine, possibly due to its lowest CL value. However, the CL value of *n*-PFHxS was not significantly lower than those of branched isomers except 2*m*-PFHxS, leading to a similar percentage of *n*-PFHxS in the three kinds of samples. As shown in Figure 3A, the predicted blood isomeric proportions of PFHxS using eq 2 were in agreement with the measured results, intensifying that the model was effective in predicting the isomeric profiles of PFASs. On the other hand, the result indicated that PFHxS mainly originated from direct exposure with a negligible contribution from its precursors.[32]

In the case of PFOA, 3*m*-PFOA was not detected in the urine samples, while other isomers were detected in all the samples (Figure 3B). The *n*-PFOA was the predominant isomer in pooled food (82.3%), blood (97.0%), and urine (94.9%), followed by *iso*-PFOA (10.3, 1.2, and 2.0%), 3*m*-PFOA (3.4, 0.7%, and not detected), and other branched isomers (<3% for each). The proportion of *n*-PFOA in the food was significantly lower than that in blood ($p < 0.001$). According to eq 2, the predicted % *n*-PFOA in the blood was 92.8%, which was significantly lower than the measured value

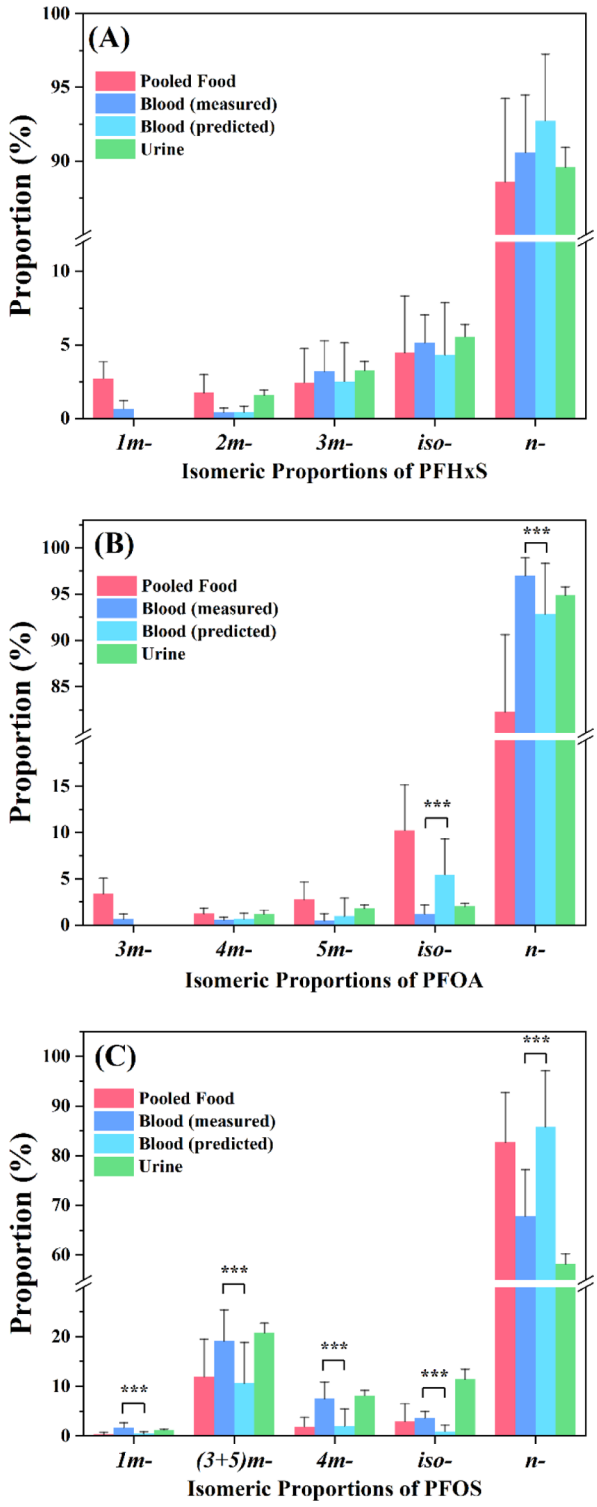

Figure 3. Isomeric profiles of (A) PFHxS, (B) PFOA, and (C) PFOS in the pooled food, blood (measured), blood (predicted) and urine (*** represented $p < 0.001$).

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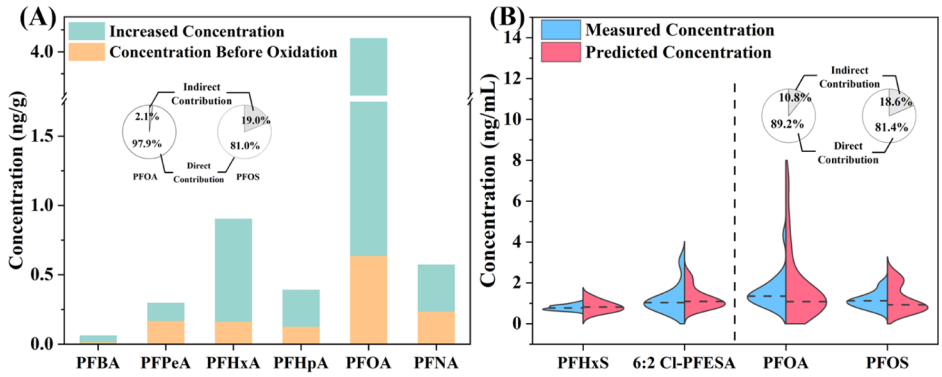

Figure 4. Quantification of indirect contribution from precursors of legacy PFASs. (A) Concentrations of PFCAs in the pooled food samples (ng/g) before and after TOP assay. Indirect contribution could be determined from external perspective by increase of PFCAs with the consideration of biotransformation rate. (B) Comparisons between the measured and predicted blood concentrations (ng/mL) of PFHxS, 6:2 Cl-PFESA, PFOA, and PFOS. The horizontal dashed line represents the median value. Indirect contribution could be calculated by the gap of measured and predicted blood concentrations from internal perspective.

(97.0%, $p < 0.01$). This finding suggested that the enrichment of *n*-PFOA in blood was not only due to its preferential accumulation over the *br*-isomers (Table S18) but also contributed by the transformation of the telomeric precursors of PFOA.[54]

For PFOS, the percentage of *n*-PFOS was dominant in the pooled food, blood, and urine, comprising 82.7, 67.9, and 58.3% of ∑PFOS, respectively (Figure 3C), followed by (3+5) *m*-PFOS (12.0, 19.2, and 20.8%, respectively). The % *n*-PFOS in blood (67.9%) was slightly lower than that in the ECF (70.0%), while significantly lower than that in the pooled food (82.7%) ($p < 0.01$). It is rather important to compare the isomer profile in blood to that in the corresponding exposure matrix, considering that general populations are exposed to PFOS through the exposure matrix but not to the ECF product. The significantly lower % *n*-PFOS in blood than in food was contradictory to the fact that the *n*-PFOS was preferentially enriched relative to the *br*-isomers.[31,55] Additionally, the lowest CL value of *n*-PFOS (Table S18) and the significantly higher predicted percentage in blood (85.8%, $p < 0.001$) also suggested that *n*-PFOS was more accumulated in human body relative to the *br*-isomers. Animal studies indicated that *br*-isomers of precursors of PFOS (PreFOSs) were preferentially transformed to the corresponding *br*-PFOS in vivo,[56,57] resulting in the apparent decline of % *n*-PFOS in blood. Thus, the significantly lower % *n*-PFOS observed in the human blood suggested that indirect exposure due to transformation of PreFOSs made a great contribution to the PFOS body burden in human.[31] The significant differences between the predicted and measured blood isomeric proportions of PFOS ($p < 0.001$ for each isomer) indicated that the participants were likely exposed to considerable amounts of precursors.

TOP Assay. As shown in Figure 4A, the mean total concentration of PFCAs in the food samples increased by 4.37 ng/g after TOP treatment, equivalent to 315% increase, suggesting that there was a large amount of unknown precursors of legacy PFAAs in the foods. PFOA exhibited the highest average increase of 554%, followed by PFHxA (457%) and PFBA (372%), implying the presence of a large amount of unknown C_8, C_6, and shorter-chained precursors.[32,58] Although both the precursors of PFOA (PreFOAs) and PreFOSs could be oxidized to PFOA, the PreFOAs mainly yield perfluoroheptanoic acid (PFHpA) rather than PFOA.[32] A relatively lower increase of PFHpA (212%) than PFOA (554%) indicated that the increased PFOA was mainly contributed by PFOS precursors. In addition, the blood % *n*-PFOS was negatively correlated with the enhanced amount of PFOA after TOP assay ($p < 0.05$), further supporting that the increased PFOA was mainly derived from PreFOSs.[32] This also intensified the fact that intake of PreFOSs might account for the low % *n*-PFOS in human blood. Furthermore, based on the method developed by Zhang et al.,[59] the contributions of PreFOSs and PreFOAs to the increased PFOA were estimated to be 84.6 and 15.4%, respectively. This suggested that a large amount of PreFOSs were present in the foods, which were subjected to in vivo transformation and served as an indirect source of PFOS in humans. It was estimated that 52.3% of PFHxA increase was contributed by the precursors of PFHxS (PreFHxSs), while the other 47.7% was contributed by the precursors of PFHxA (PreFHxAs). The results suggested that the health risk of legacy PFASs in humans could be underestimated if only their concentrations were measured in the food.

After the TOP assay, the concentrations of PFOA and PFOS in the pooled food increased by approximately 1 and 11 times, magnifying that there were indirect contributions of PFOS and PFOA human body burdens due to transformation of their precursors. To further explore the contribution due to indirect exposure, the biotransformation rate of 1.7% for PreFOAs was applied based on the compiled previous in vivo studies.[60] However, limited data were available for the biotransformation rate of PreFOSs; thus, the biotransformation rate of *N*-EtFOSA to PFOS in earthworm was applied (3.2%).[24,35] The indirect contributions of precursors to PFOS and PFOA human body burden were estimated to be 19.0 and 2.1%, respectively. The applied biotransformation rate might influence the result. However, the estimated result was comparable to the previous estimations and supported by the following internal and external exposure association assessment, suggesting the credibility of the rate we applied.[24,43]

Correlation between External and Internal Exposure Levels. The relationships between the EDIs and corresponding blood concentrations of the PFASs were inspected. Except for 6:2 Cl-PFESA, which displayed a significant correlation between its EDI and blood concentration ($p < 0.05$), no correlation was observed for PFOA, PFOS, and PFHxS (Table S19). In our study, a 1 week duplicate food was used, which

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was considerably representative of the dietary custom of the volunteers. Thus, the lack of correlations between the EDIs and human blood concentrations for PFOA, PFOS, and PFHxS might be related to contributions of their precursors.[9] Although PFHxS might have indirect contribution from precursors, the consistently measured and predicted isomeric profiles in the human blood suggested that indirect contribution of PFHxS might be minor to be taken into consideration.

Thus, the one-compartment pharmacokinetic model (eq 2) was applied to estimate the blood concentrations of 6:2 Cl-PFESA and PFHxS. If the bioavailability, F, was set to 100%, which was usually used in most of the previous studies,[21,23] the predicted blood concentrations were distinctly higher than the measured values (Figure S2). This indicated that the bioavailability (F) greatly affected the predicted blood concentration of PFASs and 100% bioavailability may lead to overestimation. Assuming that the median predicted concentrations were comparable to the measured ones, the F values would be 12.1 and 16.7% for PFHxS and 6:2 Cl-PFESA, respectively (Figures 4B and S3). As mentioned above, both 6:2 Cl-PFESA and PFHxS barely had indirect contribution from precursors. Given that the PFASs have a similar bioavailability in the food matrices, the calculated F values could be used for estimation of PFOA and PFOS blood concentrations. The F value of 6:2 Cl-PFESA (16.7%) was used for PFOS considering that they have similar chain length. The median predicted concentration of PFOS in the blood was 0.92 ng/mL, which was significantly lower than the measured concentration ($p < 0.05$). The gap between the measured and predicted concentrations could be attributed to the indirect contribution from PreFOSs. Accordingly, the indirect contribution of PreFOSs was estimated to be 18.6%, which was comparable to the result estimated by the TOP method (19.0%). With regard to PFOA, previous in vivo studies reported that the bioavailability of PFOA was lower than that of PFOS in carp and approximately 10% in mice.[27,61] As a result, the bioavailability of PFHxS (12.1%) might be more reasonable for PFOA.

The median predicted PFOA concentration was 1.23 ng/mL (two obvious outliers excluded), which was significantly lower than the measured concentration (1.38 ng/mL). The indirect contribution from PreFOAs was calculated to be 10.8%, which was higher than that estimated by the TOP method. One plausible reason might be that the PreFOSs generally belong to the N-substituted PFOSA derivatives while PreFOAs come from a variety of groups (FTSs, diPAPs, fluorotelomer carboxylic acids, perfluoroalkyl phosphonic and phosphinic acids); thus, the biotransformation rate adopted in the external exposure assessment might not be accurate due to the complex composition of PreFOAs.[2] However, the comparable indirect contribution of PFOS estimated from both methods suggested that the indirect contribution of PFOS was around 20%. The results of this study provide first line information on quantitative contribution of indirect sources to human exposure of PFOS and PFOA. The limitation of this study is the relatively small population size. Additionally, the participants had very similar characteristics. Thus, the impacts of some demographic characteristics, such as age and body mass index (BMI), on the EDI estimation could not be discovered, and more studies are warranted in future.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.1c07465.

Details on pretreatment methods, instrumental analysis conditions of HPLC-MS/MS, spiked recoveries, limit of detection and limit of quantification, equations of EDI calculation and precursor compositions, concentrations of PFASs in exposure matrices, human blood, and urine, renal clearance rates, isomer profiles in human blood, estimated daily intake, correlation analysis of PFASs concentrations in blood, and comparison of measured and predicted blood concentrations (PDF)

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Notes

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